PEDIATRIC INTENSIVE CARE UNIT

Policy Name : Blood Transfusion

Policy Number : BT – PICU 4/3 (9)

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PEDIATRIC INTENSIVE CARE UNIT

CONTENTS

- PURPOSE
- SCOPE
- ABBREVIATIONS
- DEFINITIONS
- MATERIAL AND EQUIPMENT
- POLICY STATEMENTS
- PROCEDURE AND RATIONALE
- DOCUMENTATION
- RESPONSIBILITIES
- MEASUREMENTS
- REFERENCES
- CROSS REFERENCES
- DISTRIBUTION LIST
- REVIEW SUMMARY

PEDIATRIC INTENSIVE CARE UNIT

PURPOSE:

Implementation of this policy will help to ensure that:

1. There is clear guidance to ensure that the procedure of blood transfusion is standardised in the paediatrc intensive care unit.
2. All staff act in accordance with this policy to make sure the safety of blood transfusion and prevent the transfusion related complications.

SCOPE:

First step – Verify physician's order regarding all the essential information pertaining to blood transfusion

Second step – Post transfusion monitoring

ABRREVATIONS:

PRBC	-	Packed Red Blood Cells
FFP	-	Fresh Frozen Plasma
IV	-	Intra Venous
OVR	-	Occurrence Variance Report
IPSG	-	International Patient Safety Goals

DEFINITION:

Blood transfusion:

PEDIATRIC INTENSIVE CARE UNIT

Blood transfusion is the process of transferring whole blood or blood components from donor to recipient through intravenous route for therapeutic purposes

MATERIALS / EQUIPMENT:

1. Signed consent form
2. Transfusion verification form
3. 0.9% Normal saline
4. Blood and blood product
5. Blood transfusion set
6. Infusion pump
7. Blood warmer(If necessary)
8. Tourniquet
9. Alcohol swab
10. Venous catheter and three way adaptor
11. Adhesive tape
12. Disposable glove
13. Label

POLICY STATEMENTS:

1. Written order by the physician in the physician's order sheet
2. The rate of PRBC transfusion in children is determined by assessment of the absolute Hb level, child's health status, acuteness and severity of blood loss.
3. The expected increase in the hemoglobin level is approximately 2–2.5 g/dl for each 10 ml/kg of PRBCs transfused.
4. Identification of patient by two identifiers (IPSG 1)
    4.1. Verify with the physician order for blood typing and cross matching

4.2. Prior to drawing blood for typing and cross matching
4.3. Prior to send the blood sample to the laboratory with the request
4.4. Prior to collect the blood products from the blood bank
4.5. Prior to blood administration

5. Consent must be signed by patient's parents prior to transfusion. In special cases, such as absence of parents, consent form should be signed by two physicians.
6. Cross matching result has only three days validity and specimen should be resent after 3 days for further blood transfusion.
7. Blood transfusion shall be initiated within 30 minutes after blood is taken from the Blood bank
8. Blood should not be stored in Unit Refrigerators. If unable to begin within 30 minutes, immediately return product to blood bank for safe storage
9. Check and sign the following information at patient's bedside by 2 registered nurses before infusion:
   9.1. Name of the patient
   9.2. Medical record number
   9.3. Blood serial number
   9.4. Expiration date on the blood bag label
   9.5. Blood group compatibility
   9.6. Serial number on the request and on the right wrist of patient
10. Standard precautions are to be followed when handling blood products.
11. A standard blood transfusion set must be used for transfusion. Do not use the ordinary IV tubing set.
12. All packed red cell and FFP transfusions are administered via a volumetric infusion pump.
13. Blood transfusion is to begin slowly and should be completed within four hours. FFP over 2 hours and platelet is administered rapidly over 30 minutes.
14. Blood should only be warmed in a blood warmer. All the machines seem to be set at 37 degree Celsius. Blood should never be warmed in a bowl of hot water as this could lead to hemolysis of the red cells which could be life-threatening when transfused.
15. Nurse should observe patient closely for the first 5 to 15 minutes after the blood transfusion is initiated.

16. Closely observe patient throughout the procedure for signs and symptoms of fever, allergy or anaphylactic reaction.
17. Crystalloid solution other than 0.9% saline and all medications are incompatible with blood products as it leads to agglutination and transfusion reaction.
18. Do not transfuse platelets through a giving set that has previously been used for red cells or other blood component as this may cause aggregation and retention of platelets in the line.
19. Blood transfusion should be stopped and physician should be notified immediately if signs of blood transfusion reaction occur.

PROCEDURE:

S. No	Procedure	Rationale
Preparatory phase:		
1.	Verify physician's order for following : 1.1. Indications for blood transfusion 1.2. Blood product type 1.3. Number of units to be transfused 1.4. Rate of transfusion 1.5. Transfusion time 1.6. Have any accompanying	A written order requesting the blood transfusion therapy must be made by a physician prior to implementation of the procedure

PEDIATRIC INTENSIVE CARE UNIT

	Medications: e.g. Furosemide	
2.	Identify the patient by using two identifiers (Name and medical record number – first verification)	Correct identification of patient as per IPSG goal 1
3.	Assess the patient with special focus of history of blood transfusion or reaction	Provide baseline data
4.	Check blood return for venous access or insert new intravenous line	Provide a safe and patent access for blood transfusion
5.	Ensure that the procedure and desired outcome was explained to patient's family, if present	Help to allay anxiety and get family cooperation
6.	Check blood transfusion consent has been taken and completed properly	Informed consent provides legal protection
7.	Complete the initial assessment by checking vital signs	Helps to inform the physician, if there is hyperpyrexia or any other abnormal assessment
8.	Complete pre blood extraction patient identification by two nurses (second Verification)	Correct identification of patient as per IPSG goal 1
9.	Obtain blood sample, preferably selecting a large vein and uses large bore cannula	Allow free flow of transfusion and also some degree of mobility
10	Places cross matching printed number on the following:	Avoid errors in blood transfusion

	10.1. On the order sheet 10.2. On the sample tube 10.3. On the blood request forms 10.4. On the blood transfusion forms 10.5. On the patient's right wrist	
11	Complete the blood transfusion request form with required information and send the request to the blood bank completing all the formalities	To ensure the accuracy of blood and blood products
12	Obtain blood product by registered nurse from blood bank	To ensure the accuracy and safety of blood and blood products
13	Collects blood products from the blood bank completing the formalities as per hospital policy and checks the following information: 13.1. Name of the patient 13.2. Medical record number 13.3. Blood serial number 13.4. Expiration date on the blood bag label 13.5. Blood group compatibility	To prevent error in order to safe delivery of blood products to the patient
14	Maintain universal precaution while handling blood and blood products	To prevent infection

PEDIATRIC INTENSIVE CARE UNIT

15	Ensures that the blood administration set (with double chamber and an integral mesh filter) primed with normal saline ensuring that blood tubing set is free from air bubbles.	Blood products are only compatible with normal saline 0.9%. Flushing prevents infusion of air and potential air embolism
16	Double check the blood bag with patient's medical record number before the transfusion preferably charge nurse (Third verification)	To prevent error in order to safe delivery of blood products to the patient
17	Complete visual checking of the blood bag for following: 17.1. Pack integrity 17.2. Frothiness 17.3. Cloudiness 17.4. Presence of large clots sediments 17.5. Evidence of hemolysis in the plasma If present, returns to blood bank	To prevent complications due to damaged blood products
18	Verify blood products at the bedside in front of the patient with another nurse (Final verification)	To prevent error in order to safe delivery of blood products to the patient
19	Ensure that emergency drugs such as hydrocortisone, antihistamine and adrenaline are kept available	To manage any transfusion reactions effectively

PEDIATRIC INTENSIVE CARE UNIT

	Performance phase:	
20	Perform hands hygiene and wear sterile gloves.	To maintain sterile technique.
21	Invert blood component bag gently 2-3 times	Equally distribute cells throughout the preservative solutions.
22	Spike the product bag using aseptic technique with clamp close. Opens the blood transfusion set clamp and allows the blood to fill the filter chamber.	Aseptic technique decreases chances for infection
23	Start transfusion and remains bedside for close monitoring at least 15 minutes	most of the adverse reactions may occur during this time
24	If no signs of reaction for the first 15 minutes, regulate the flow rate according to Physician's order to run at least 2 to 4 hours (drop factor of blood transfusion is 10 drops/ml)	Maintaining the prescribed rate of flow decreases the risk of fluid volume excess while restoring vascular volume
25	Monitor pre-transfusion vital signs then 5 minutes after blood transfusion has started and follow with 10 minutes , 15 minutes, first hour, second hour, third hour and fourth hour after completion and records in the blood transfusion form.	Frequent monitoring of vital signs will help nurse alert quickly to any transfusion reaction

PEDIATRIC INTENSIVE CARE UNIT

26	Observe for chills, flushing , dyspnea, rashes or other signs of transfusion reaction	Assess for transfusion reactions
27	In case of transfusion reaction is noted, do following:	
	27.1. Stop the infusion immediately and notifies the charge nurse or head nurse	Prompt intervention may minimize potential for serious complications
	27.2. Reassure the patient	To reduce anxiety of patient
	27.3. Maintain venous access (Keep line patent with normal saline)	To maintain patency of IV line
	27.4. Ensure that treating physician and blood bank is notified	To manage effectively To provide baseline data
	27.5. Check vital signs and complete quick assessment for consciousness, breathing patterns or rashes	To resuscitate patient effectively
	27.6. Manage effectively with all emergency resuscitation measures if it appears life threatening	To identify any clerical errors
	27.7. Double check for any clerical errors such as patient	

	identification on identification band, request form and on the blood bag 27.8. Send the following to the blood bank for rechecking (freshly collected blood and urine sample from the patient, discontinued blood bag and tubing in yellow bag and copy of completed transfusion reaction form) 27.9. Monitor vital signs and keep close watch on patient 27.10. Documents in nurse's note 27.11. Complete and submit an OVR form with physician's comments	To investigate the reason for transfusion reaction To identify deterioration of health at early stage Documentation ensures accountability and continuity of care
28	Upon completion of transfusion flush IV cannula with normal saline.	Keep IV line patent
	Follow up phase:	
29	Discard the waste tubing and other material into appropriate waste bags	Proper disposal of waste materials
30	Complete the blood transfusion form and attach in file	Documentation helps in future reference

31	Document the completion of procedure in nurse's notes and update intake and output chart.	Documentation ensures accountability and continuity of care

DOCUMENTATION:

1. Physician order should be written with all needed information clearly and legibly.

2. Complete the blood transfusion form and document the completion of procedure in nurse's notes based on the following aspects:
    1. Type of blood products transfused
    2. Number of units transfused
    3. Amount of blood transfused
    4. Length of time taken
    5. Outcome of transfusion
    6. Any adverse reactions
    7. Patient's tolerance for transfusion

RESPONSIBILITIES:

Physician provides a written order for type of blood product, amount, duration of transfusion with his or her signature and stamp.

Clinical Instructor is responsible to provide knowledge based competency lectures for the staff and ensure that the nurse is completing the competency skills check off in the skill station and in the units. Update the knowledge of staff and supervise the actual performance in the unit.

Head Nurse is responsible in ensuring all staff are competent to perform blood transfusion and implement the physician order.

PEDIATRIC INTENSIVE CARE UNIT

Charge Nurses are responsible in double checking the physician's order, process of blood transfusion and monitoring actual performance of staff nurse. Provide continuous supervision and incidental guidance for bed side staff.

Staff Nurses are responsible in completing the KBCL and skill check off for blood transfusion and attending an in-service education conducted by nursing administration. Staff should be competent enough to perform the procedure as per the policy statement and should follow the steps as instructed in policies.

MEASUREMENT:

1. 100% of the staff is supposed to know the procedure of blood transfusion and comply with it.
2. 100% of the staff should have to complete successfully the KBCL and competency check off

REFERENCES:

1. Ministry of Health, Policy and Procedure Manual, KSA
2. Lippincott Manual of Nursing Practice, 11th edition by Nettina
3. Nurse's Guide to Clinical Procedure, 6th edition by Temple and Johnson

CROSS REFERENCES:

1. Blood bank policy
2. Nursing administration policy

DISTRIBUTION LIST:

1. Nursing administration
2. Quality department

REVIEW SUMMARY:

1. None